

# Substation to Storage

## A feasibility thesis

Repurposing decommissioned Czech transformer stations (transformovny / rozvodny) as grid batteries with rooftop solar — instead of demolishing them

Quantitative model on realised Czech market data · June 2026

### Verdict — a conditional yes, for the right sites

Reusing a decommissioned medium- or high-voltage substation's grid connection for a battery (plus rooftop PV on the surrounding buildings) adds about EUR 6.0m–EUR 9.8m of net present value over an equivalent greenfield battery in our representative 20 MW / 40 MWh case — almost entirely from avoided grid-connection capex and roughly two years' earlier operation. Crucially, that advantage survives even if balancing-service revenue collapses to zero. But it applies only where the retained connection is at least a few MW at distribution or transmission voltage; the thousands of small 0.4/22 kV kiosk trafostanice are far too small. And because the connection belongs to the grid operator, the realistic developer is the DSO/TSO or a partner holding a connection agreement — exactly who is already doing it at Chvaletice, Kladno and Vitkovice.

Scope & method: a representative-site techno-economic model, not site due diligence. Day-ahead arbitrage is computed from realised 2019-2026 Czech auction prices (the same LP battery backtest used elsewhere in this project); balancing revenue, capex and the connection saving are parameterised from 2025-26 Czech market data with explicit scenarios and sensitivities. All figures are illustrative and not investment advice.

# 1 The premise and the question

Across Czechia, ageing transformer and switching stations (*transformovny* and *rozvodny*) are reaching the end of their life — a distribution substation is typically rebuilt every 30-40 years, and older or redundant ones are demolished outright. The question posed here: rather than demolish, could such a site host a grid battery — its cables and connection are already there — with solar panels on the surrounding buildings, and is that worth doing?

The short answer is that the buried value of these sites is almost never the building or the old iron; it is the **grid connection**. Whether reuse beats demolition is therefore a question about one number: how many megawatts, at what voltage, the connection can still carry.

# 2 What is actually being demolished — and which sites matter

'Transformer station' spans four orders of magnitude of capacity. The feasibility verdict flips across that range, so the taxonomy is the heart of the analysis.

| Tier | Typical site                              | Voltage / size       | Battery it supports | Verdict          |
|------|-------------------------------------------|----------------------|---------------------|------------------|
| A    | Kiosk / pole <i>trafostanice</i>          | 0.4/22 kV, <1 MVA    | < 0.5 MW            | No — too small   |
| B    | Distribution substation                   | 22-110 kV, 10-40 MVA | 5-20 MW             | Yes — sweet spot |
| C    | Transmission node / ex-thermal switchyard | 110-400 kV, 100s MW  | 50-300 MW           | Ideal            |

The thousands of small kiosk *trafostanice* (Tier A) being swapped out in network upgrades are the wrong target: a battery there would be a fraction of a megawatt, useful at most for a local community microgrid. The prize is Tier B and C — and at the top of the range sit retired coal and industrial sites whose switchyards already carry hundreds of megawatts. One caveat: the common Czech pattern is to rebuild the substation in place (as at Trebic-Ptacov, demolished while its replacement runs alongside), so the connection often stays in network use. 'Free connection' is real only where genuine spare capacity remains.

# 3 Why the grid connection is the prize

A grid connection is the longest-lead, scarcest and often costliest part of a battery project. In benchmark all-in BESS capex of roughly \$125/kWh outside China, about \$50/kWh — some 40% — is installation and connection, not cells. In Germany the transmission connection queue is effectively full; the UK is reforming connections explicitly to clear the battery pipeline; across Europe developers are pivoting to 'co-location' precisely to share an existing connection. A live connection means skipping a two-to-four-year queue and saving on the order of EUR 60-250 per kW of capacity.

## The lever, in one line

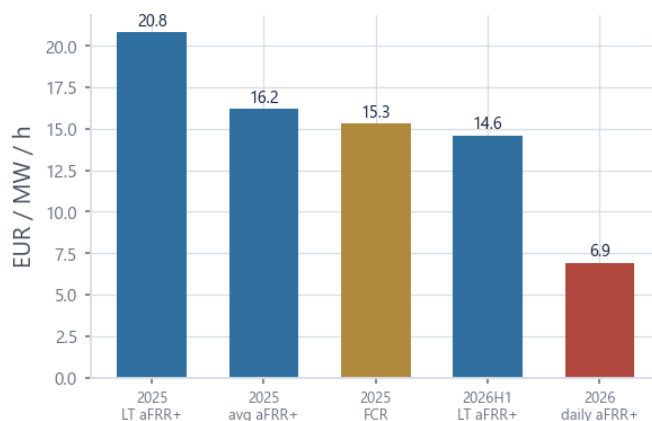
A retired substation hands a battery developer the two things money cannot quickly buy: a connection agreement that already exists, and the years that would otherwise be spent waiting for one.

# 4 The Czech market right now

The timing is favourable on capacity and policy, and treacherous on revenue.

**Boom and policy.** Czech battery capacity is set to rise from about 2.3 GWh today toward 6 GWh by the end of the decade. The 2026 Energy Act amendments let batteries trade wholesale and provide FCR, aFRR and mFRR, and streamlined grid-connection procedures; the Modernisation Fund and RES+ co-fund storage. Tellingly, the flagship projects all sit on reused industrial or coal connections: the 230 MWh Chvaletice battery on a lignite power-station site, the 90 MWh Kladno project, CEZ's battery in the Vitkovice industrial complex, and BESS Lipnice in the Sokolov coal region. The thesis is not hypothetical — it is the prevailing build pattern.

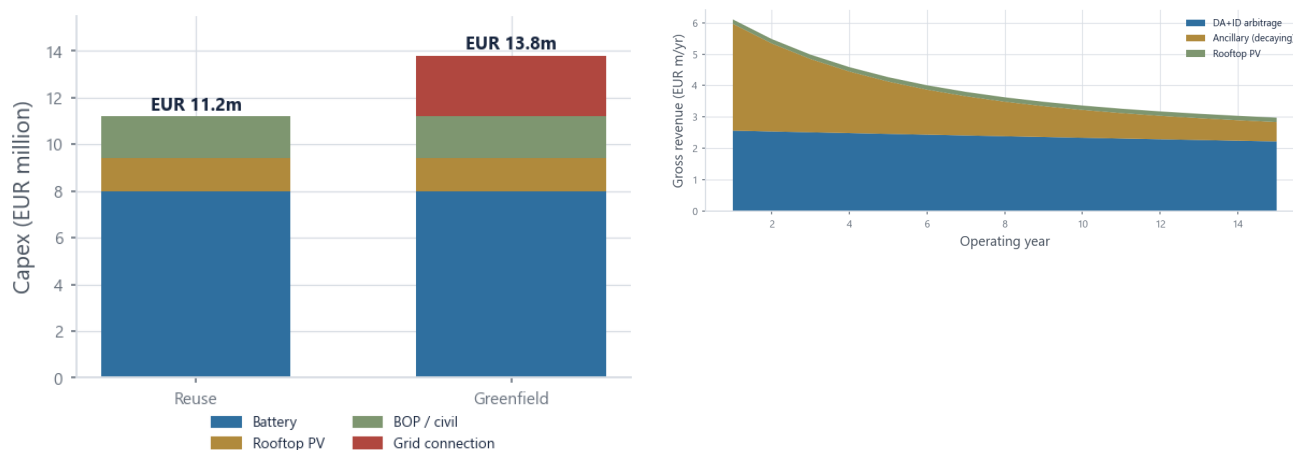
**The revenue trap.** But the easy money is disappearing. The whole Czech FCR-plus-aFRR requirement is only around 370 MW, and the fleet is flooding it: aFRR+ long-term prices fell about 34% in 2025 (to ~16 EUR/MW/h) and FCR about 40% (to ~15 EUR/MW/h), with daily aFRR+ down to ~7 EUR/MW/h by mid-2026. Commentators call it 'the end of the illusion of easy returns.' A top operator still earned around EUR 340k/MW in 2025 via fully-automated multi-market trading, but that figure is falling and will not survive saturation. The durable floor is energy arbitrage — and here Czechia is unusually rich: its day-ahead spreads are wide and widening, worth about EUR 88k per MW per year on day-ahead alone in 2025 in our LP backtest, before intraday.



Czech ■ EPS balancing-service prices, EUR/MW/h — the collapse that pushes battery economics toward arbitrage.

## 5 A quantitative feasibility model

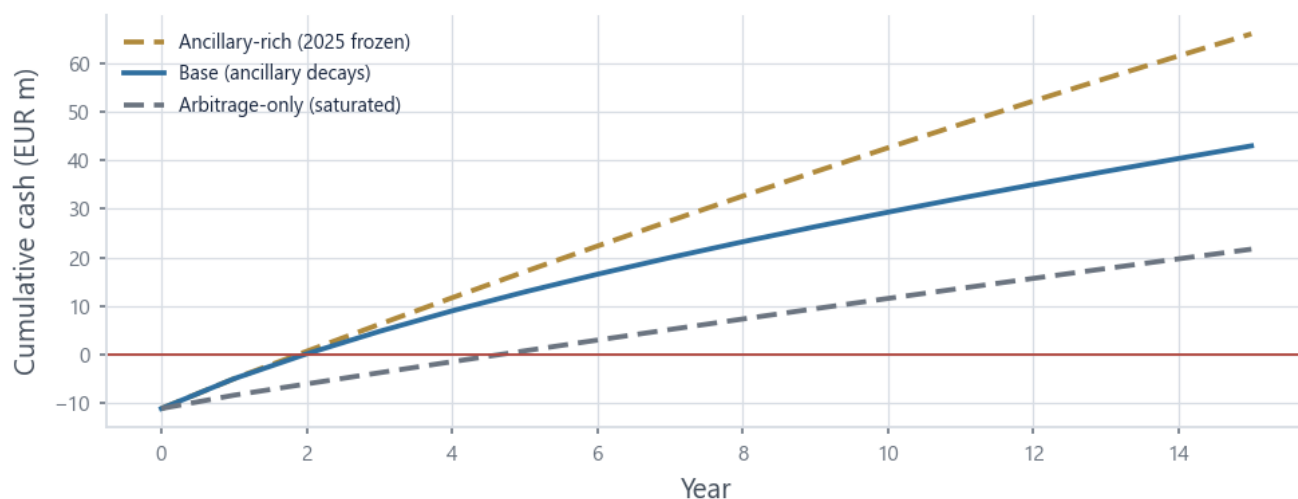
We model a representative Tier B site: a retired 110/22 kV substation with about 20 MW of usable retained connection, fitted with a 20 MW / 40 MWh (2-hour) battery and 2 MWp of rooftop PV on the surrounding buildings. Revenue stacks day-ahead and intraday arbitrage (from realised prices), a decaying balancing-service contract, and solar. We compare reuse against an identical greenfield battery that must pay for a new connection and wait 2 years to energise.



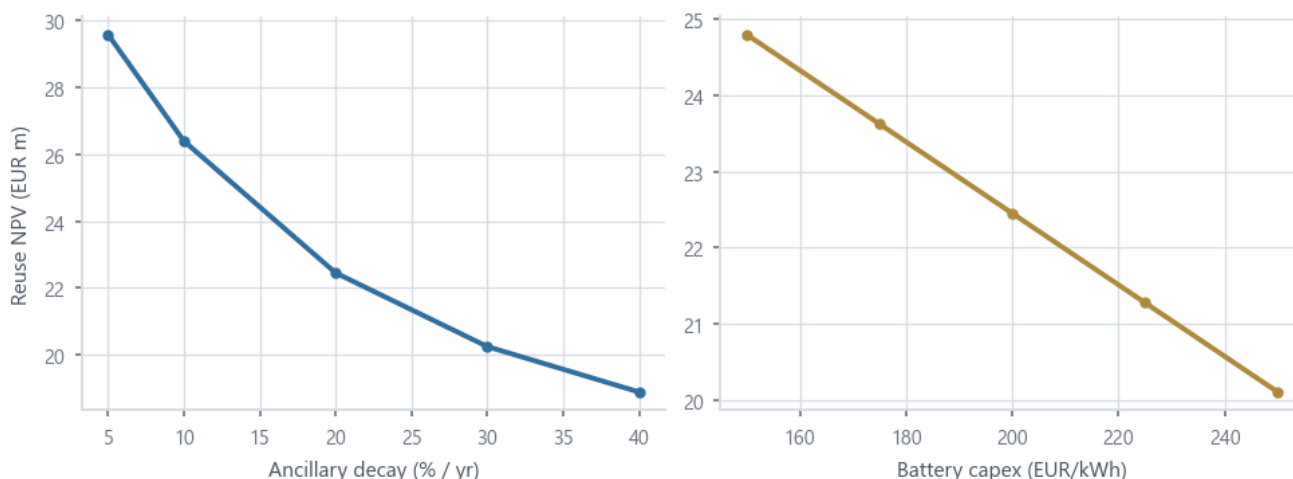
Left: capex — reuse avoids the grid-connection block entirely. Right: the gross revenue stack in the base case, with balancing revenue decaying into an arbitrage-and-solar floor.

### Results — reuse / greenfield, 15-year, 8% discount.

| Scenario                     | NPV EURm    | IRR       | Payback | Reuse edge EURm |
|------------------------------|-------------|-----------|---------|-----------------|
| Ancillary-rich (2025 frozen) | 34.1 / 24.4 | 51% / 24% | 2y / 5y | +9.8            |
| Base (ancillary decays)      | 22.5 / 14.4 | 44% / 20% | 2y / 5y | +8.1            |
| Arbitrage-only (saturated)   | 8.0 / 2.0   | 19% / 10% | 5y / 9y | +6.0            |



Cumulative cash for the reuse case across scenarios. Even arbitrage-only — balancing revenue gone — pays back inside the asset life.



Sensitivity of reuse NPV to the two swing variables: how fast balancing prices decay, and battery capex per kWh. The project stays NPV-positive across the plausible range; the reuse edge over greenfield is more robust still, because it is a capex and timing difference, not a revenue bet.

## 6 Rooftop solar on the surrounding buildings

Solar is a sensible companion but a junior partner. Roof area on a substation's ancillary buildings is limited, so a couple of MWp is realistic — perhaps EUR 0.1-0.2m a year of generation at a cannibalised midday capture price. Its real value is synergy and zero marginal connection cost: the PV charges the battery cheaply at the midday solar trough (often near or below zero — Czechia saw a sharp rise in negative-price hours), which the battery then shifts into the evening peak, and both assets share the one connection and one set of balance-of-plant. It improves the economics at the margin; it does not carry the project.

## 7 When it is worth doing — the decision rule

- **Connection test.** Reuse beats demolition when the retained connection carries at least roughly 5 MW at 22 kV or above and real spare capacity exists. Below that, the connection is worth less than the cleared land — demolish.
- **Owner test.** The connection is the DSO's or TSO's asset. The credible developer is the grid operator, the site owner, or a partner with a connection agreement — not an outside party assuming the cables are free for the taking.
- **Condition test.** The value is the line, the bay, the land and the permits — not the old transformer, which a battery replaces with its own inverters and transformer anyway. Brownfield liabilities (contamination, asbestos, foundations) must net out below the connection saving.
- **Counter-incentive.** Demolition itself can be subsidised (the Ministry for Regional Development funds brownfield clearance), and the operator may need the bay for the replacement substation. Reuse must clear that bar too.

## 8 Risks

- **Balancing saturation.** Modelled explicitly; the tiny ~370 MW national requirement saturates quickly and prices are already in steep decline.
- **Arbitrage cannibalisation.** The same battery build-out that saturates balancing also narrows the day-ahead spreads arbitrage relies on — so even the arbitrage-only case is optimistic over a full 15 years; treat it as an upper bound on the floor.
- **Bankability.** Standalone merchant batteries are reported at 5-7% unlevered IRR in tougher European markets; Czech spreads are better but volatile, and revenue is exposed, not contracted.

- **Brownfield and technical.** Contamination, protection upgrades and connection re-rating can erode the saving; due diligence per site is essential.

## 9 Verdict

### Worth doing — selectively, and as the grid owner

For Tier B and C sites, reusing the connection for a battery beats both greenfield and demolition: the connection is the scarce asset, and reuse banks its value plus two years of lead time, an edge of roughly EUR 8.1m of NPV in the base case that holds even if balancing revenue disappears. It is not a strategy for the small kiosk stations, and it is realistically executed by the grid operator or a connection-holding partner — which is precisely the pattern now playing out across the Czech coal and industrial fleet. The headline revenue numbers will fall as the market saturates; the connection-reuse advantage will not.

### References

- CzechTrade / ess-news — Czech large-scale storage; AlphaESS Chvaletice (230 MWh) and Kladno (90 MWh); CEZ ESCO Vitkovice battery; SUAS BESS Lipnice.
- oEnergetice / Prumyslova ekologie — Czech balancing-service auction results 2025-26 ('the end of the illusion of easy returns'); ■EPS FCR/aFRR/mFRR prices.
- ess-news (2026) — 'Making the case for brownfield battery builds'; co-location and connection-reuse economics in Europe.
- Ember / SolarPower Europe — European BESS capex and revenue-stacking benchmarks 2025-26.
- solarninovinky / FENIX, Nano Energies, Suená — Czech and German operator revenue per MW, 2025.
- This project — CZ day-ahead LP arbitrage backtest on realised 2019-2026 EPEX prices (Energy-Charts / Fraunhofer ISE).

## Appendix The feasibility model (code)

The full techno-economic model used above. Assumptions are named constants with their sources in comments; the three scenarios and the sensitivity sweep are the output.

python · analysis/substation\_battery.py

```
"""
Substation-to-storage feasibility model.

Thesis: a decommissioned MV/HV transformer substation ("transformační /
rozvodná stanice") that is slated for demolition retains the one thing a
new battery project waits years and pays the most for – a live grid
connection at meaningful capacity. Reusing it for a battery (optionally
with rooftop PV on the surrounding buildings) can beat both a greenfield
battery and plain demolition. This model quantifies it for a
representative site and stress-tests the result.

Revenue is built bottom-up:
- Day-ahead arbitrage: taken from the LP battery backtest on realised
  CZ auction prices (results/battery_daily_CZ.csv) – the hard,
  data-grounded number.
- Intraday uplift: a documented multiplier on DA-only revenue.
- Ancillary services (FCR/aFRR/mFRR): a contracted-capacity adder that
  DECAYS, reflecting the 2025-26 collapse in Czech balancing prices and
  the tiny national requirement (~370 MW of FCR+aFRR) saturating as the
  fleet grows from ~2.3 to ~6 GWh.
- Rooftop PV on surrounding buildings: generation x a cannibalised
  capture price; secondary to the battery, with a self-charge synergy.

The reuse case differs from greenfield by (a) avoided grid-connection
capex and (b) ~2 years earlier commercial operation – both pure upside.

Every assumption is a named constant with its source in the comment, and
the three scenarios plus the sensitivity sweep are the real output: the
verdict is conditional, not a single number.
"""

from __future__ import annotations

import json
import sys
from dataclasses import dataclass, asdict, field
from pathlib import Path

import numpy as np
import pandas as pd

sys.path.insert(0, str(Path(__file__).parent.parent))

RES = Path(__file__).parent.parent / "results"
OUT = RES

# ----- economics -----
@dataclass
class SiteConfig:
    # --- representative "Tier B" site: a retired 110/22 kV distribution
    #   substation with ~20 MW of usable retained connection capacity ---
    power_mw: float = 20.0
    duration_h: float = 2.0          # 2-hour system -> 40 MWh
    pv_mwp: float = 2.0             # rooftop PV on surrounding buildings

    # --- capex (EUR) ---
    capex_batt_eur_per_kwh: float = 200.0  # turnkey LFP 2026, ~EUR170-240/kWh
    capex_pv_eur_per_kwp: float = 700.0    # CZ commercial rooftop
    capex_bop_eur_per_kw: float = 90.0     # integration/civil at a reused site
    # the lever: greenfield grid connection avoided by reuse (EUR per kW)
    connection_capex_eur_per_kw: float = 130.0
    demolition_avoided_eur: float = 400_000.0  # partial; connection bay retained

    # --- year-1 revenue (EUR per MW of power) ---
    arb_da_eur_per_mw: float = 88_400.0    # CZ 2025 DA-only LP (data-grounded)
    intraday_uplift: float = 0.45          # ID on top of DA-only
    ancillary_eur_per_mw_y1: float = 170_000.0  # 2025 contracted FCR/aFRR/mFRR blend
    ancillary_floor_eur_per_mw: float = 35_000.0
    ancillary_decay: float = 0.20          # per year toward the floor

    # --- PV revenue ---
    pv_yield_mwh_per_mwp: float = 1_000.0  # CZ ~950-1050
    pv_capture_eur_per_mwh: float = 70.0    # cannibalised solar-hour capture

    # --- opex & lifetime ---
    opex_pct_of_capex: float = 0.02
    reserved_capacity_eur_per_mw_yr: float = 6_000.0  # ongoing DSO charge proxy
    revenue_degradation: float = 0.02      # per-year derate (fade/augmentation)
```

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arb_growth: float = 0.01          # mild structural widening then flat
life_years: int = 15
discount_rate: float = 0.08
greenfield_delay_years: int = 2    # connection queue avoided by reuse

@property
def energy_mwh(self) -> float:
    return self.power_mw * self.duration_h

def _capex(cfg: SiteConfig, reuse: bool) -> dict:
    batt = cfg.energy_mwh * 1000 * cfg.capex_batt_eur_per_kwh
    pv = cfg.pv_mwp * 1000 * cfg.capex_pv_eur_per_kwp
    bop = cfg.power_mw * 1000 * cfg.capex_bop_eur_per_kw
    conn = cfg.power_mw * 1000 * cfg.connection_capex_eur_per_kw
    total = batt + pv + bop + (0.0 if reuse else conn)
    return {"battery": batt, "pv": pv, "bop": bop,
            "connection": (0.0 if reuse else conn), "total": total}

def _annual_revenue(cfg: SiteConfig, year: int, scenario: str) -> dict:
    """Year-indexed revenue (year 0 = first operating year)."""
    arb = (cfg.arb_da_eur_per_mw * (1 + cfg.intraday_uplift)
           * (1 + cfg.arb_growth) ** year) * cfg.power_mw
    if scenario == "ancillary_rich":
        anc_unit = cfg.ancillary_eur_per_mw_y1          # frozen 2025
    elif scenario == "arbitrage_only":
        anc_unit = 0.0
    else: # base: decay toward floor
        anc_unit = (cfg.ancillary_floor_eur_per_mw
                    + (cfg.ancillary_eur_per_mw_y1 - cfg.ancillary_floor_eur_per_mw)
                     * (1 - cfg.ancillary_decay) ** year)
    anc = anc_unit * cfg.power_mw
    pv = cfg.pv_mwp * cfg.pv_yield_mwh_per_mwp * cfg.pv_capture_eur_per_mwh
    gross = (arb + anc + pv) * (1 - cfg.revenue_degradation) ** year
    return {"arbitrage": arb * (1 - cfg.revenue_degradation) ** year,
            "ancillary": anc * (1 - cfg.revenue_degradation) ** year,
            "solar": pv, "gross": gross}

def _npv(rate: float, cashflows: list[float]) -> float:
    return float(sum(cf / (1 + rate) ** t for t, cf in enumerate(cashflows)))

def _irr(cashflows: list[float]) -> float | None:
    lo, hi = -0.9, 2.0
    f_lo, f_hi = _npv(lo, cashflows), _npv(hi, cashflows)
    if f_lo * f_hi > 0:
        return None
    for _ in range(200):
        mid = (lo + hi) / 2
        f = _npv(mid, cashflows)
        if abs(f) < 1.0:
            return mid
        if f_lo * f > 0:
            lo, f_lo = mid, f
        else:
            hi = mid
    return (lo + hi) / 2

def cashflow(cfg: SiteConfig, scenario: str, reuse: bool) -> pd.DataFrame:
    cap = _capex(cfg, reuse)
    rows = []
    # greenfield pays capex now but earns nothing during the connection queue
    delay = 0 if reuse else cfg.greenfield_delay_years
    opex = cap["total"] * cfg.opex_pct_of_capex + \
           cfg.reserved_capacity_eur_per_mw_yr * cfg.power_mw
    # year 0 = build/spend; operations begin after `delay`
    horizon = cfg.life_years + delay
    for t in range(horizon + 1):
        if t == 0:
            net = -cap["total"]
            rev = {"gross": 0.0, "arbitrage": 0.0, "ancillary": 0.0, "solar": 0.0}
        elif t <= delay:
            net = 0.0
            rev = {"gross": 0.0, "arbitrage": 0.0, "ancillary": 0.0, "solar": 0.0}
        else:
            op_year = t - delay - 1
            rev = _annual_revenue(cfg, op_year, scenario)
            credit = cfg.demolition_avoided_eur if (reuse and t == 1) else 0.0
            net = rev["gross"] - opex + credit
            rows.append({"year": t, "net": net, **rev})
    df = pd.DataFrame(rows).set_index("year")
    return df

def evaluate(cfg: SiteConfig, scenario: str) -> dict:

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out = {}
for reuse in (True, False):
    df = cashflow(cfg, scenario, reuse)
    cfs = df["net"].tolist()
    npv = _npv(cfg.discount_rate, cfs)
    irr = _irr(cfs)
    cum = np.cumsum(cfs)
    payback = next((i for i, v in enumerate(cum) if v >= 0), None)
    cap = _capex(cfg, reuse)
    out["reuse"] if reuse else "greenfield" = {
        "capex_total": round(cap["total"]),
        "capex_connection": round(cap["connection"]),
        "npv": round(npv),
        "irr": round(irr, 4) if irr is not None else None,
        "payback_years": payback,
        "y1_gross_revenue": round(df.loc[df.index > 0, "gross"].iloc[0]) if (df["gross"] > 0).any() else 0,
    }
out["reuse_advantage_npv"] = round(out["reuse"]["npv"] - out["greenfield"]["npv"])
return out

def sensitivity(cfg: SiteConfig) -> dict:
    base = evaluate(cfg, "base")["reuse"]["npv"]
    grid = {}
    sweeps = {
        "capex_batt_eur_per_kwh": [150, 175, 200, 225, 250],
        "ancillary_decay": [0.05, 0.10, 0.20, 0.30, 0.40],
        "connection_capex_eur_per_kw": [60, 100, 130, 180, 250],
        "discount_rate": [0.06, 0.08, 0.10, 0.12],
    }
    for field_name, vals in sweeps.items():
        grid[field_name] = []
        for v in vals:
            c = SiteConfig(**{**asdict(cfg), field_name: v})
            grid[field_name].append({"value": v,
                                     "npv": round(evaluate(c, "base")["reuse"]["npv"])})
    return {"base_npv": round(base), "sweeps": grid}

def load_arb_from_backtest() -> float | None:
    """Use the realised CZ 2025 DA-only LP revenue if available."""
    fp = RES / "battery_daily_CZ.csv"
    if not fp.exists():
        return None
    df = pd.read_csv(fp, index_col=0)
    df.index = pd.to_datetime(df.index, utc=True)
    y2025 = df[df.index.year == 2025]
    return float(y2025.revenue.sum()) if len(y2025) else None

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